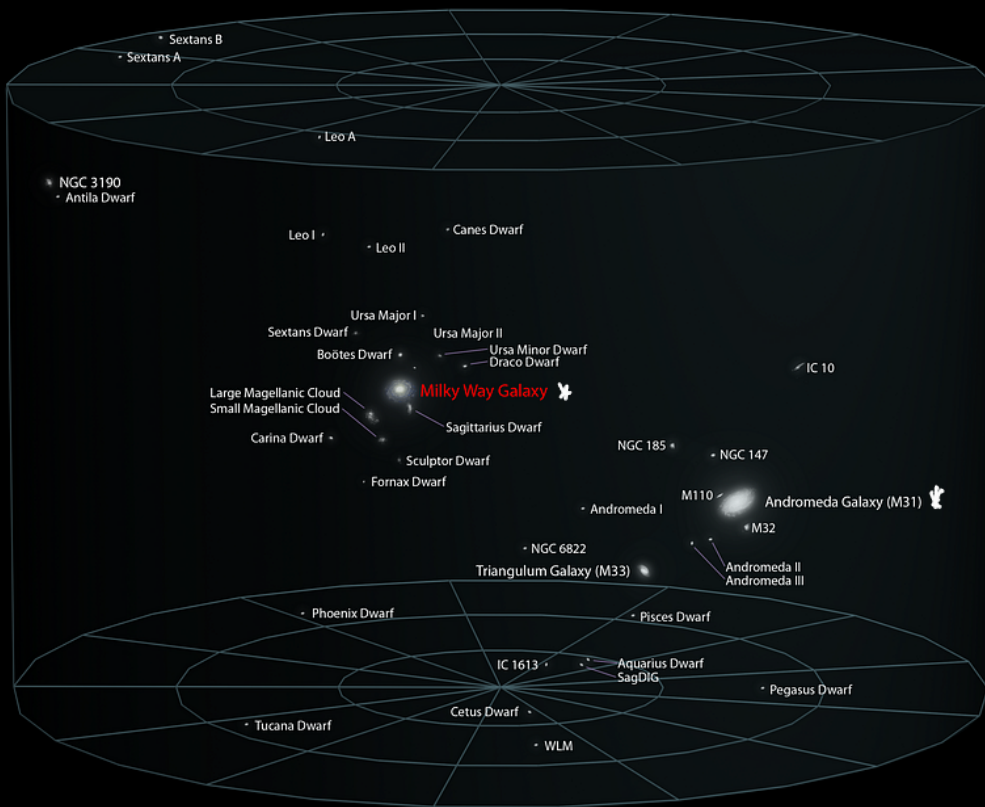
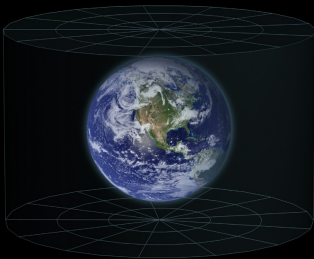


Local Galactic Group

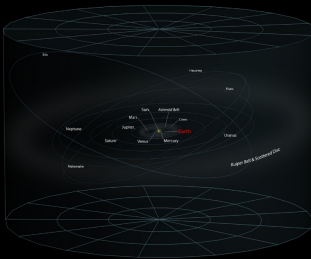


40

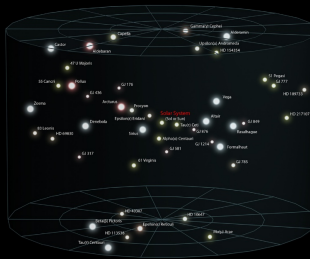
Earth



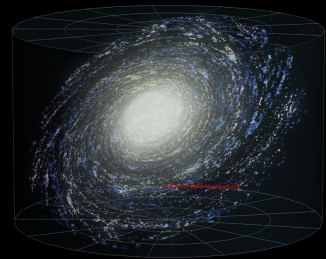
Solar System



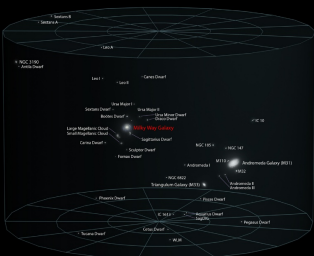
Solar Interstellar Neighborhood



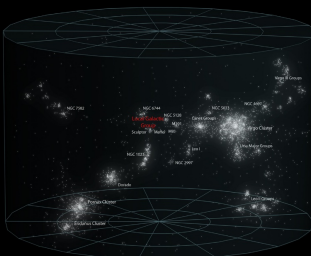
Milky Way Galaxy



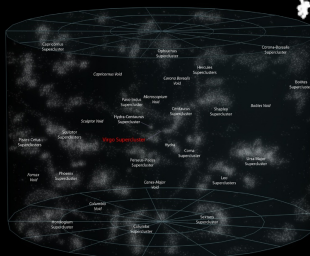
Local Galactic Group



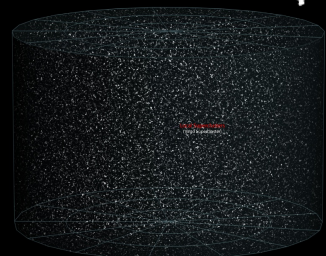
Virgo Supercluster



Local Superclusters



Observable Universe



local group : $D \approx 1 \text{ Mpc}$

$V_{\text{pec}} \approx H_0 D + v_{\text{pec}}$

• MW + M31 : Dominant

• Gravitationally bound

• Err. + Dwarf.

→ All have neg. Pec. velocities

Nearby Group: $D \approx 5-15 \text{ Mpc}$

• Brightest in our nbrhd.

• Contains other systems similar to local group

Virgo Cluster: $16 \sim 100 \text{ Mpc}$

• 16 Mpc from us

• ~ 250 large galaxies / ~ 2000 smaller

• Much more dense: Very grav. bound

• All gal here still have pec. vel.

→ Sits in the local supercluster



30 Mpc

Coma Cluster: $\approx 100 \text{ Mpc}$

• Redshifts start to rise

$200 \text{ Mpc} \approx cz \quad z \ll 1$

$V \approx cz$

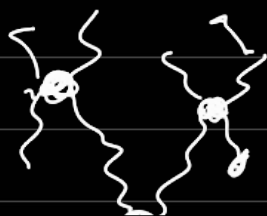
• Great Wall: Spans $\sim 70 \text{ Mpc}$

$$* N_{\text{gal}} = \int_{L_1}^{L_2} \phi(L) dL \quad | \quad L_{\text{tot}} = \int L \phi(L) dL *$$

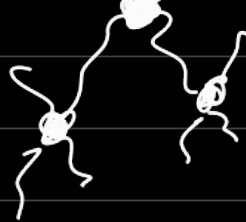
• Start seeing a pattern:

→ Galaxies are not randomly dist.

→ "Cosmic web structure" (Pg 80 of notes)



• local group sits out in a web



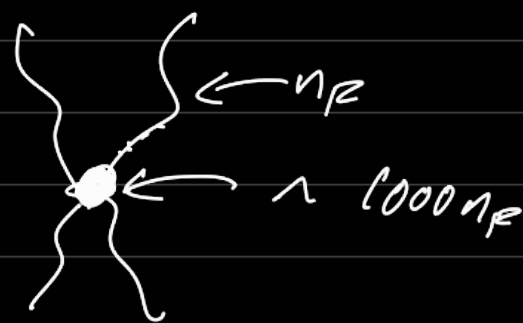
- * Empty voids $\sim 35 \text{ Mpc}$
- * "faded" sectors $\sim 20 \text{ Mpc}$
- * Dense central regions

* Scales larger than 140 Mpc , Universe looks

- { (i) Homogeneous }
- { (ii) Isotropic }

Clusters of galaxies:

- Filament: $n_F = (n_F)_{\text{gal}}$
- Center: $\sim 1000 n_F$



(i) Regular Clusters

- Spherical
- ~ 1000 bright gal ← large
- Very few spirals
- Coma: ex

* Don't find any spirals within central dense areas

(ii) Irr. Clusters

- Irr shape
- Smaller ~ 100 bright gal
- More spirals
- ex: Virgo cluster

Morphologies:

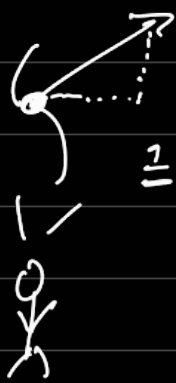
- Clusters: often dominated by a few large ell. galaxies
- A lot: mergers
- CD: cluster Dominant ←

Physics:

- Similar to ellipticals: no structure
- Dispersion vel. $\sigma^2 = \langle V_{gal}^2 - \langle V_{gal} \rangle^2 \rangle$
 V_{gal} : vel of a gal. $\uparrow$

Observations: $\tilde{\sigma}_r = 350 - 1000$ km/s

σ_r : line-of-sight vel. dis



$$\tilde{\sigma}_r^2 = 3 \sigma_r^2 \Rightarrow \boxed{\tilde{\sigma}_r = \sqrt{3} \sigma_r}$$

Irregular:

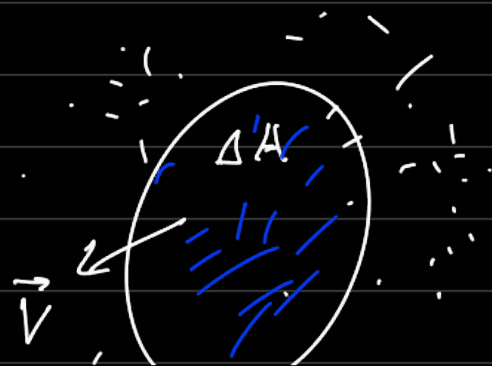
- Have a large X-ray emission
- Second largest X-ray source besides ABNs
 - Come from Intra-Cluster Medium (ICM)
- $ICM \approx T_{ICM} \approx 10^7 - 8$ K
- * Temp comes from Bremsstrahlung emission
 - Brem $\leftrightarrow$ Free-Free emission
- * A continuum spectrum from ICM
 - Fe lines
- * → $ICM [Fe] \sim 0.3 - 0.5 Z_\odot$
- ICM is not primordial

Ram pressure stripping:

$$\vec{F} = \frac{d\vec{p}}{dt} \approx \frac{\Delta m}{\Delta t} \vec{v}$$

- Assumes $\vec{v}$ constant

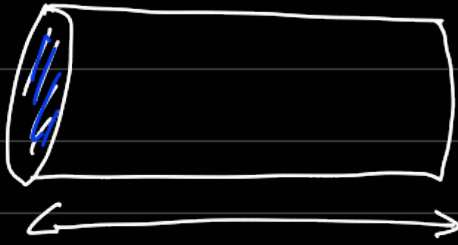
$$\vec{p} = \vec{F} \Delta t = \Delta m \vec{v}$$



$$\frac{\Delta A}{\Delta A} \left(\frac{\Delta A}{\Delta t} \right)^2$$

$$\star \left(\frac{\Delta M}{\Delta t} \right) = \Delta A v \rho_{\text{ICM}} \star \quad \left| \quad \rho \sim \frac{\Delta M}{\text{(Vol)}} = \frac{\Delta M}{\Delta A v \Delta t} \right.$$

$$\bullet \text{ Vol} = \Delta A L = \Delta A v \Delta t \quad \curvearrowright$$



$$v = \frac{L}{\Delta t}$$

replace $\Delta M / \Delta t \xrightarrow{L} \underline{P} = \frac{\Delta A v \rho_{\text{ICM}} v}{\Delta A}$

$$\boxed{P = \rho_{\text{ICM}} v^2}$$

